A statistical analysis of NYC subway ridership

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Abstract

With millions of people living in urban regions with public transportation, it is important to evaluate the systems in order to optimize for a multitude of factors. Subways, roadways, and busses are relied upon for daily commuting. I aim to understand the factors that effect daily ridership on New York City's (NYC) subway, a large facet of the Metropolitan Transportation Authority (MTA). I combine open source weather data and ridership data with a two statistical models to undergo an empirical study. I find statistically significant and compelling results then subsequently use robust checks evaluate their validity.

1 Introduction

With 472 subway stations and over 200 bus routes, the Metropolitan Transportation Authority (MTA) in New York City (NYC) is the largest transportation network in the United States. Due to its existence in the greater ecosystem of the city, the MTA is both impacted by and impacts a variety of external factors, such as economic opportunity through labor mobility, social connectivity, and vitality of businesses. It is important to understand policies and factors with relation to the MTA because transportation impacts real people's day to day activities and well being.

Motivation to increase understandings of substitution effects across transportation systems within a locality also exists due to the potential additional price policy changes in other transportation modes in NYC. One of recently elected NYC Mayor Mamdani's election platforms increasing affordability of living in NYC, with one method being the proposed elimination of bus fares.[1] If the increased marginal cost of driving into the CBD pushes consumers to use the subway, then there could also exist a causal impact of decreasing the cost of using the busses as a method of transportation. Thus, understanding these effects is important as more efficient allocations of resources and more informed anticipations of demand, could improve both the city and those living in it.

1.1 Background

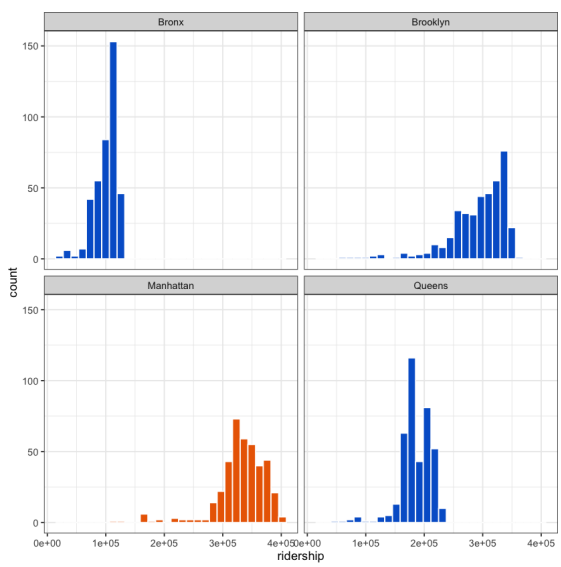
In January 2025, the City of New York levied $9 tax on drivers coming into the Central Business District, henceforth referred to as the CBD, located in lower Manhattan, during peak hours, 5am-9pm. The tax amount depended on time of day, payment method (EZ-Pass, toll by mail, etc), and vehicle type.[2] Though a primary of the implementation of this tax was for traffic reduction in Manhattan, other goals included improving the air quality

[1] AP News
[2] NYC MTA Agency Info

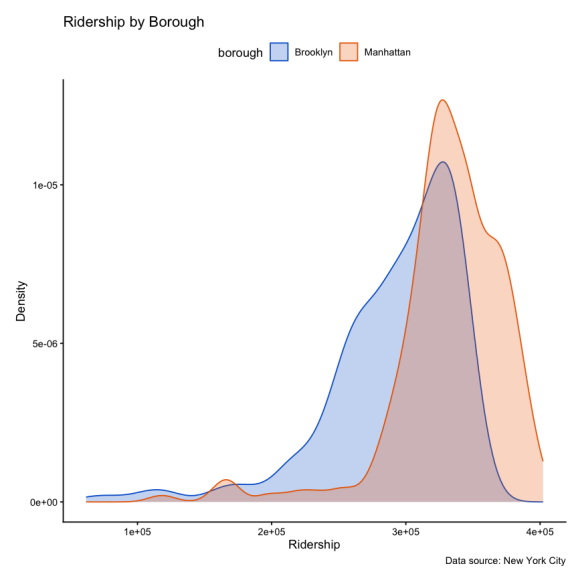
within the CBD and the generation of revenue for improvements across MTA infrastructure. The congestion tax will be expanded in the coming years and the tax levied will increase incrementally.

Research and analysis has been published on the outcomes. Baghestani and Tayarani find that in the first few months of the pricing, the air quality was improved, as seen with the microscopic emissions measures.[3] The City of New York measured air quality change as well, with monitors in the CBD, including in Midtown and at Manhattan terminus of the Williamsburg Bridge, and outside, in Bronx and Staten Island. They conclude that there is a steady or decreasing levels of find particle air pollution.[4]

The study most relevent to my paper is an economic analysis of the short run effects of the congestion pricing. This uses credit card transaction data, estimated foot traffic, and air quality data to understand the multidimensional effects of the tax. Using a synthetic control, in addition to increased traffic speeds within the CBD, the researchers find that welfare, measured by individual value, is increased more outside the CBD than inside.[5]



Note: Each borough has different frequent values for ridership



Note: Manhattan is more likely to have a day where ridership is above 400,000

[3] Baghestani and Tayarani, 2020.
[4] NYC Environmental Health Data Portal
[5] Cook and Kreidieh, 2026.

1.2 Data

Paramount to the evaluation of the NYC MTA was open source data collected and maintained by the MTA as part of their open data program under the New York City Transit and Staten Island Railway organizations. This data is collected to facilitate a quantitative and objective evaluation of the city's transportation and is published on an open platform, maintained by data.ny.gov, to foster transparency and civic engagement. Available datasets in this repository include data regarding key performance indicators, customer feedback, and ridership across different modalities.

For this analysis on the subway ridership, two main datasets were leveraged, the *MTA Subway Hourly Ridership: 2024-2024*,[6] and the *MTA Subway Hourly Ridership: Beginning 2025*.[7] These datasets share the same attributes, with the only difference being collection before or after 2025.[8] Observations in this dataset are hourly counts of entrances through turnstiles, reported each hour by each of the over 400 subway stations, excluding those in Staten Island. Variables included in this data includes the hour, the unique station identification number, borough, ridership, and class of fare payment.

This analysis relies on the valid collection of the data. The count data is not inclusive of people who ride the subway without going through the turnstile. While we cannot capture the full potential substitution effect from driving to subway utilization, we can focus instead on the count of people who do comply with the turnstile. There exists MTA data on the estimated subway fare evasions rate, which could be utilized in further analysis.

Supplementary data included information regarding temperature and precipitation. Weather data was sourced from the National Centers for Environmental Information, NOAA. [9]. This includes measures of temperature, rain presence, and snowfall observed from the JFK airport.

[6] MTA-Subway-Hourly-Ridership-2020-2024

[7] MTA-Subway-Hourly-Ridership-2020-2024

[8] This does introduce another source of invalidity if the measurement process changed when the policy of study is implemented

[9] NOAA open source data

1.3 Approach and Tools

This report focuses exclusively on understanding daily subway ridership, either at a specific station, at a borough level, or with a difference-in-differences approach. Ticket prices remained constant.

This paper attempts to predict the variance in day to day subway utilization using a multiple linear regression. It is imperative to reflect on how the models themselves as well. Commentary regarding model and variable selection, interpretation of diagnostic plots, and the reporting of basic model evaluation metrics is in the appendix.

Finally, to understand the potential association between a per unit roadway tax and subway ridership, I use a causal framework with borough level groups.

Did the infliction of the NYC congestion tax for passageways leading into the CBD have a direct result on the transportation decisions made?

Connecting a theory about the real world to a question requires that that question be answerable.[10] We can transform the question above into a more tangible question: "Did ridership before and after the tax being levied change with respect to borough?" This question is interesting as there are many New Yorkers who travel into Manhattan for work or other daily activities from outside other parts of the city or state. Did this tax impact their daily routine?

It is of crucial note that this serves only as a framework or a way to learn how the processes works. Due to the presence of invalidity, any apparent result is superficial, and does not adequately represent the truth. However, it can be argued that sometimes frameworks and ideas on how a process could potentiallly by carried out are interesting. Even if the results were insignificant but the set up was valid, I would still keep the section in order to combat the file drawer effect because maybe the insignificant can hold meaning as well.

[10] The Effect Book, Chapter 2.

2 Methods and Results

2.1 Multiple Linear Regression

I use a multiple linear regression (MLR) to understand the associations between ridership and other variables due to its interpretability. However, although ridership is a numeric variable, it is also discrete and nonnegative. The is also time related data, so errors are likely to be correlated.

Regression is performed on a single station's metrics is done to provide insight with noise reduction and on the city overall, excluding Staten Island, to understand overall patterns. Assumptions of the MLR include linearity, homoskedasticity, independence, and normality. There are large violations of these assumptions for all models of both levels.

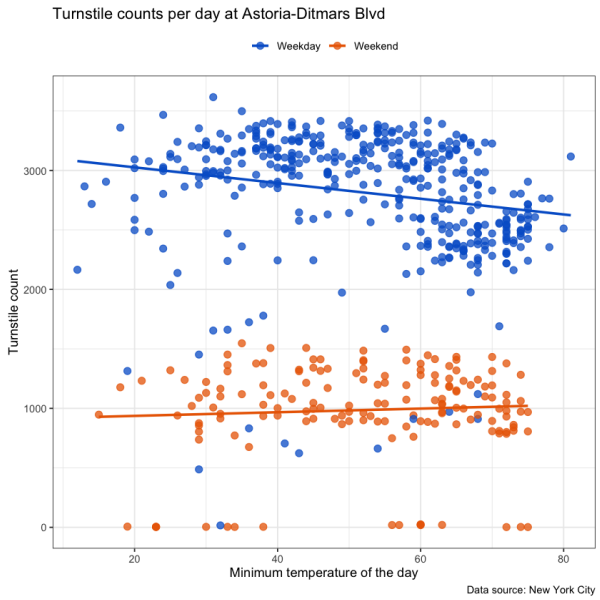
For both datasets in which the regressions are used, I use various model selection and evaluation methods. These are included in the appendix. Most notably, I train the model on a subset of the data and generate the mean squared prediction error from comparing the values to tuples not used for the estimation of parameters.

The parameter estimates could be useful in predicting high ridership days, thus allowing for proactivity, especially at the single station. The estimates could also be relevant at the city level for administrators to understand the difference that various factors could have on the amount of use the subway system is getting overall. This could allow for planning or program adjustments as well to optimize affordability and customer experience.

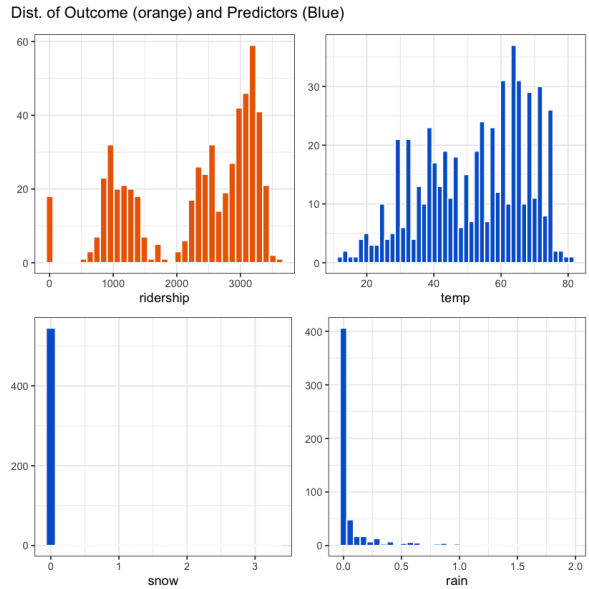
2.1.1 Astoria

I hone in on the Astoria-Ditmars Blvd (N,W) station,[11] located in the Astoria neighborhood of Queens, for June 2024-December 2025. I join the outcome variable, daily ridership, with independent variables from the JFK airport weather data.

[11] Representative station within Queens selected randomly



(a) Daily turnstile counts at a subway station in Queens, NYC



(b) Exploration of Astoria Data

Figure 2: Astoria data visualizations

Left: Ridership is lower on weekends regardless of temperature. Right: Snow and rain variables are skewed.

The results for the model fit with the complete data at the station level can be seen here Table 1. The following linear model specification corresponds to the fourth column in the below regression.

$$y = \beta_0 + \beta_1 x_1 + \delta_1 x_2 + \delta_2 x_3 + \beta_3 (x_1 \times x_3) + \beta_4 x_1^2 + \varepsilon \tag{1}$$

where x_1 = temperature, x_2 = rain (boolean), and x_3 = weekend (boolean). All other specifications are reductions of this model.

These assumptions do not hold for the models fit. Thus, the parameters are not the best linear unbiased estimators. See appendix for diagnostic plots and elaboration (Figure 6). Even after the outcome variable was transformed, there were still violations. Thus, this is a biased model with low explanatory or predictive power.

Table 1: Regression Results for Astoria data

	ridership			
	(1)	(2)	(3)	(4)
temp	-4.284^{***}	-4.278^{***}	47.672^{***}	45.010^{***}
	(1.318)	(1.323)	(8.240)	(8.246)
Weekend	$-1{,}827.404^{***}$	$-1{,}827.271^{***}$	$-1{,}828.217^{***}$	$-2{,}236.352^{***}$
	(45.577)	(45.676)	(44.119)	(153.301)
rain_d		-2.512	-9.156	-2.610
		(43.865)	(42.382)	(42.192)
temp^2			-0.526^{***}	-0.522^{***}
			(0.082)	(0.082)
Temp × Weekend				7.812^{***}
				(2.812)
Constant	$3{,}036.566^{***}$	$3{,}037.066^{***}$	$1{,}890.377^{***}$	$2{,}014.956^{***}$
	(73.230)	(73.814)	(193.308)	(197.305)
Observations	557	557	557	557
R^2	0.745	0.745	0.762	0.766
Adjusted R^2	0.744	0.743	0.761	0.763
Resid. Std. Error	485.785	486.223	469.646	466.813
Resid. Std. Error df	554	553	552	551
F Statistic	808.119^{***}	537.778^{***}	442.491^{***}	359.846^{***}
F Statistic df	2; 554	3; 553	4; 552	5; 551

***p<0.01, **p<0.05, *p<0.1

Note: The boolean variable for rain is not significant for any model

All of the models have a statistically significant F statistic at the .05 significance level. Models 1 and 2 report a negative impact of the temperature, in Fahrenheit, on daily ridership, both with statistically significant coefficients around $\hat{\beta}_1 = -4.2$. This contrasts sharply with the models with more parameters, models 3 and 4, as they both report a positive association between the temperature and ridership, with a greater magnitude than the smaller models, $\hat{\beta}_1 > 45$. The marginal impact of an increase in temperature (in Fahrenheit) is seen in the addition of 45 more people entering the Astoria subway station that day.

The more profound and consistent result is the association between the weekend indicator and daily ridership. On average, with all else equal, if the day is a Saturday and Sunday, there are 1,800 people less who enter the Astoria station. This could maybe point to the

fact that the subway is a method of transportation to work, which often occurs during the weekday.

Comparisons of the MSPE, AIC, and BIC of the models are included in the appendix. (Appendix subsubsection A.1.1)

2.1.2 Borough Level

Harboring too heavy a focus on one station moves the city overall into an substandard periphery, so I broadened my view to include city wide data, minus Staten-Island. This observatiosn window is between Jun 2024-Dec 2025, inclusive of weekends, with daily aggregation including ridership during non rush hours (same as the Astoria regression above).

I use fixed effects to account for baseline borough level differences with the allowance of different intercepts for each borough for the outcome variable.

Model 4 from the regression in Table 2 is represented by this linear equation.

$$y = \beta_0 + \beta_1 x_1 + \delta_1 x_2 + \delta_2 x_3 + \beta_3 (x_1 \times x_3) + \beta_4 x_1^2 + \varepsilon \tag{2}$$

where x_1 = temperature, x_2 = rain (boolean), x_3 = weekend (boolean). x_4 accommodates is a vector of dummy variables for the boroughs to represent the baseline differences, where the Bronx serves as the baseline and Staten Island remains excluded. All other specifications are reductions of this model. Discussion on selection, diagnostics, and evaluation is included in the appendix Figure A.1.1.

Partial F-tests show that the coefficient on the dichotomous variable representing any rainfall is not statistically different than zero. This is also alluded to by both the lack of stars next to the coefficient (with the problem of multiple comparisons in mind) and the lack of an increase in adj R^2 as move from model 3 to four, where we add that variable (along with the interaction term of temp and weekend).

Table 2: Regression Results for the City

	ridership			
	(1)	(2)	(3)	(4)
temp	−694.049	7,662.618**	7,597.892***	7,498.474***
	(496.756)	(3,199.986)	(843.851)	(848.580)
Weekend	−241,728.300***	−241,743.200***	−242,292.300***	−267,716.700***
	(17,133.830)	(17,110.850)	(4,508.232)	(15,759.410)
rain	−8,807.306	−9,768.538		−9,387.935**
	(16,465.920)	(16,447.860)		(4,339.660)
$temp^2$		−84.620***	−84.200***	−84.409***
		(32.012)	(8.444)	(8.436)
Brooklyn			463,025.900***	463,025.900***
			(5,766.526)	(5,758.717)
Manhattan			946,223.700***	946,223.700***
			(5,766.526)	(5,758.717)
Queens			218,908.000***	218,908.000***
			(5,766.526)	(5,758.717)
temp × weekend				496.301*
				(288.543)
Constant	737,413.100***	552,830.700***	144,809.000***	153,636.900***
	(27,701.960)	(75,107.460)	(20,125.100)	(20,613.560)
Observations	2,232	2,232	2,232	2,232
R^2	0.083	0.086	0.936	0.937
Adjusted R^2	0.082	0.085	0.936	0.936
Resid. Std. Error	365,535.500	365,045.300	96,319.970	96,189.530
Resid. Std. Error df	2228	2227	2225	2223
F Statistic	67.549***	52.545***	5,463.581***	4,109.812***
F Statistic df	3; 2228	4; 2227	6; 2225	8; 2223

***p<0.01, **p<0.05, *p<0.1

Note: We obtain the same adj.R2 for model 4 as we do without including the rain boolean or the interaction term

Similar to that in the subsection of data in the above section, the weekend is the most prominent covariate. In all models, it is reported as statistically significant at the 0.01 level. Similar as well is the negative relationship between the weekend and the ridership associated with that day, exemplified in model 3, $\hat{\beta}_2 = -242,292.3$. A potential mechanism for a quarter of a million people less taking the subway on a weekend is that maybe people do not have to commute.

Controlling for Borough through binary indicators for Brooklyn, Manhattan, and Queens, with Bronx being the baseline, allows us to get within borough variation. I did not consider geographic area or amount of stations within each borough. The F test shows that the varying intercepts are significant.[12] Again, the rain dummy variable is not significant and does not add any predictive power to the model.

One application of this knowledge to the real world is that the difference in mean ridership across boroughs, with all other variables being 0, shows that policies aimed applied to the whole city may have a different magnitude of effects based on borough. The boroughs are not uniform, so large scale events impacting the whole cities subway system may have a larger impact on the higher ridership boroughs.

2.2 Difference in Differences

Is there evidence that the imposition of a $9 tax on drivers entering the CBD during peak hours results in a substitution effect, as seen with an increase in subway ridership into the CBD? Is the evidence convincing overall?

"A theory that cannot be tested has no value": I aim to test the applicability of the substitution theory on transit as best as I can through robustness checks. I do this by imposing a question: "Did the congestion tax in NYC cause a change in out of CBD ridership relative to inner CDB ridership?" If there is a false positive, this paper also aims to understand the limitations of its own study.

"Lack of evidence is not evidence of a lack of"[13]: I do not claim to have isolated the causal impact or modeled a direct increase in subway ridership from Brooklyn into Manhattan. However, this substitution effect could still have occurred in real life, even if undetected by my modeling.

[12] The Effect Book, Chapter 16.
[13] Dr. Zaharatos

The answer to this question, or any insight regarding it, can be useful in further iterations of this tax, or similar taxes, either in New York City or in potential implementations in other cities. The correct answer to this question could allow the effectiveness of the tax to be understood with respect to the counterfactual. If the tax was successful in reducing ingress into the CDB, then we might observe a changed level of ridership in our control group as compared to the treated group.

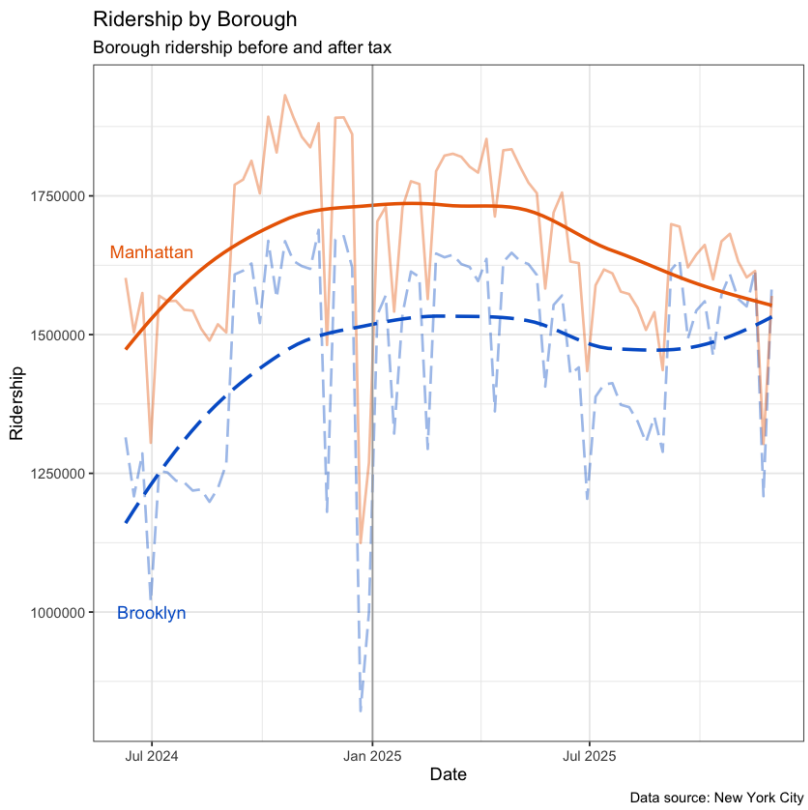
2.2.1 Design and Assumptions

As my focus was on seeing if the people who would have driven into Manhattan for work switched to public transportation as a result of this policy change, I narrowed the hourly data to focus only on ridership occurring during rush hour, or 7am-11am on weekdays. While I use casual methods, I intentionally avoid asserting causal claims.

I set Manhattan as the treatment group as the borders of the CDB are in that borough, and Brooklyn as the control group because that borough had similar levels and trends of ridership prior to the tax. Other considerations for the formulation of these groups included other geographic options, such as a isolating specific turnstile observations or focusing on areas within certain coordinate defined zones, or temporal options, such as Manhattan in prior years compared to Manhattan post tax. The simplicity of this design is justified by the comparable and necessary feasibility, but more discussion on future ideas is here Section 3.

Recognizing the credible threats to the validity is vital to responsible modeling. Randomization was not available as the assignment of the treatment was a directed and intentional intervention in one area of the city. There are oblivious spillover effect concerns. Additionally, the aggregation within the groups could pose a threat wherein people in Brooklyn could be only traveling within Brooklyn and not into the CDB, thus it seem as people are seeking an alternative to driving but are really traveling within Brooklyn or to other boroughs. It is also of note that the CBD is only in lower Manhattan (south of 60th st) whereas the whole

Figure 3: Figure Title



Note: The overall borough trends before Jan 2025 appear similar, but different after Jan 2025

borough is group into the treatment.

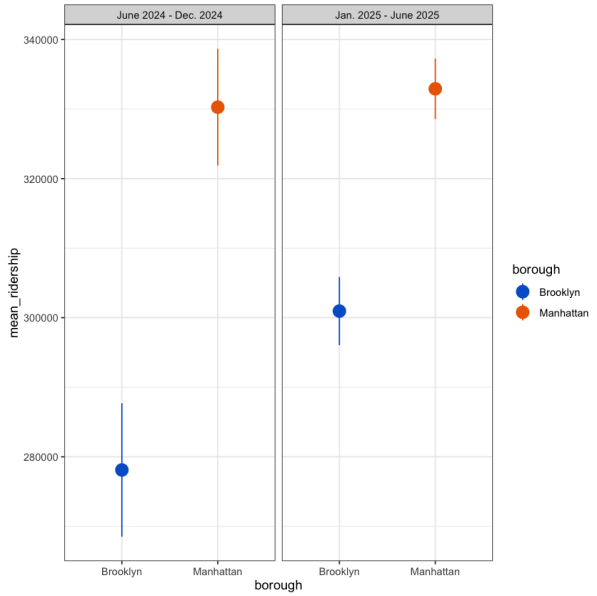
We assume a variety of factors and situations hold. We assume that the tax is salient for those would be impacted. Otherwise any treatment effect would be capturing chance rather than an intentional change in consumer behaviour. We also assume that people did not change their commuting habits too far prior in anticipation of the policy.

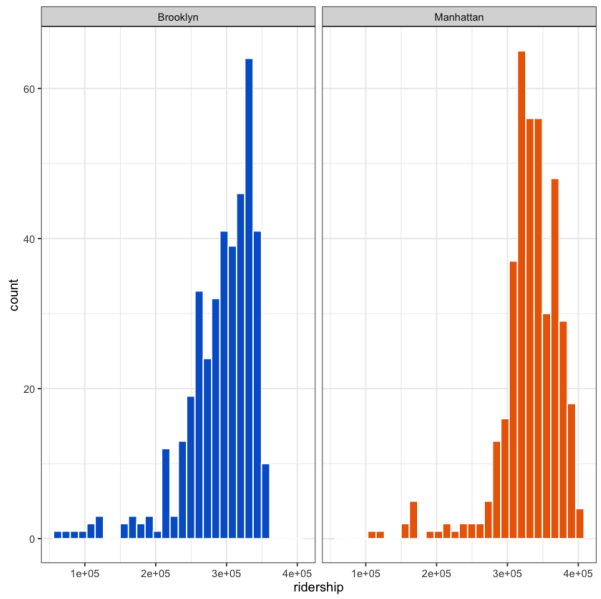
The most pertinent assumption in the difference in differences method is the imposition of the parallel trends assumption, wherein under this assumption, we could assert that the treatment group would have maintained a similar trend as the other group after the policy, had the policy not been implemented. I test this by through the assignment of placebo treatments. Sub-section A.2 If either of these pretend situations yeild a significant treatment effect, and we know there was no real treatment, then the significance of the original, non-

placebo is suspect.

While the prior trends are visually similar between groups, as seen in Figure 3, the placebo tests show that the real treatment's significance does not matter.

But we push on with a cognizance of limitations. I utilize two way fixed effects to see if there was an interaction between time (before and after the policy) and group (Manhattan vs Brooklyn). Results for the original treatment and placebo tests are here Table 3.





Note: Brooklyn's 6 month mean increased more than Manhattans

2.2.2 DiD Results

The two way fixed effects model shows that the treatment effect on ridership is -20206, and is statistically significant at the 0.01 level. This suggests that there was a difference in the change in ridership overtime, a span including the real treatment period, by borough. Relative to Manhattan's change, Brooklyn's change in ridership was larger. Overall, we would think that there is a difference between the groups. After the treatment date of January 1, 2025, the overall trend in Manhattan seems to have decreased while the Bronx maintained an increase. If we imagined a superimposed counterfactual trend line for Manhattan, daily

ridership in Manhattan would be much higher in Dec. 2025 than either Brooklyn or real Manhattan.

However, the placebo tests also yielded similar statistically significant results. A fake treatment time of Sept. 1, 2024 was run. Its significance could be, compared to the real time, that the "treatment" effect does not depend on the real time the policy was implemented. Queens was assigned as a fake treatment group, where I imagined that there was a tax to enter Queens. This result is also significant, thus suggesting that the treatment is not unique to the real treatment group of Manhattan.

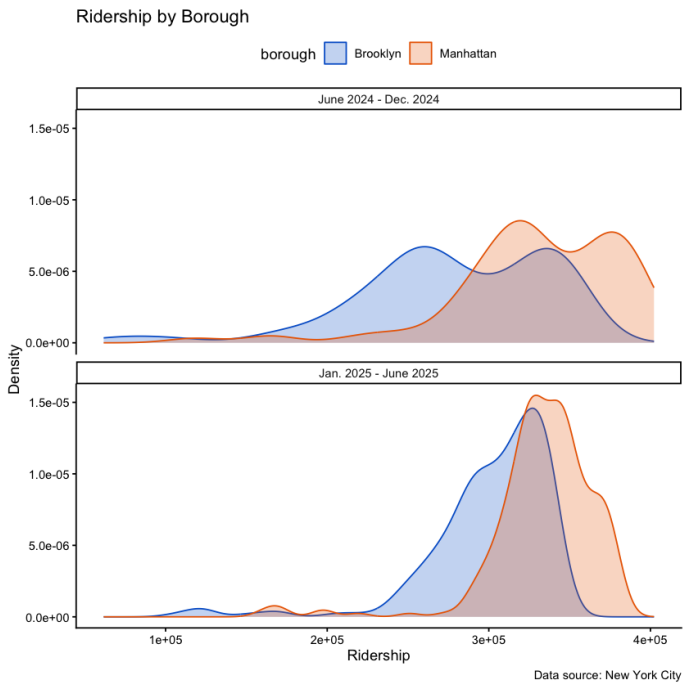
While the reported treatment effect for the real occurrence (real time and real group) appears significant, it may not be capturing the true, as shown with the significance of the fake treatment effects. These placebo tests show that the design is not robust.

Table 3: Regression Results

	Real	Placebo Time	Placebo Borough
Manhattan x Treated	-20206.267***		
	(1447.835)		
Manhattan x Fake Period		-24166.208***	
		(1983.934)	
Queens x Treated			-25607.514***
			(2644.938)
Num.Obs.	794	794	794
R2	0.981	0.979	0.963
R2 Adj.	0.961	0.958	0.926
RMSE	7002.25	7295.09	12791.88
FE: borough	X	X	X
FE: date	X	X	X

Note: Describe the table above.

Figure 4: Ridership Density by borough, faceted by time period



Note: Brooklyn appears to align closer with Manhattan after the treatment than it did before

3 Conclusions

3.1 Extension

More questions:

- Is there a substitution from single occupancy vehicles to bus use?
- What other outcomes (safety or feedback) can we predict using ridership data?
- Would the use of other methods, explanatory or predictory, show similar results?

The further work on this project can be categorized into the use of additional data sources or the use of different explanatory methods. This would increase the validity, robustness, and comprehensiveness of this study. Future questions and ideas come from the potential use of new datasets and methods.

Other data, serving as different outcomes additional predictors, could be incorporated as

well for both explanatory or predictive purpose, and begs more questions. The estimated rate of fare evasion are available and could be used to increase validity. Instances of vandalism, crime, and subway failure could be interesting outcomes. Other predictors could include economic data for the adjustment for more city level factors. Ridership counts are also published for bus lines, which could be integrated or replace the outcome variable. A question could lie in if there was a difference in the impact of the roadway tax on different categories of riders with respect to price discrimination (student discount pricing, elderly pricing, etc) or bundle pricing.

The largest improvement, with respect to data sources, in finding the truth that this project sook to find arguably lies in the availability of origin destination data. This data set estimates the amount of riders that took certain paths, entering or exiting at certain stations. This could be relevant for a causal analysis because it would allow us to see if there was a change in the amount of people who entered lower Manhattan from outside of the CBD.

Other vast improvements can be made in the methods used. While I used a basic difference in difference design with treatment and control groups at the borough level, there are many improvements that could allow for a potential real causal analysis to be undertook. For example, a more robust control group could be found in a synthesis of other cities, similar to the aforementioned study. Even if the observations came from the sole city, a better definition of groups could be at the turnstile level. Other approaches could include decreasing the level of aggregation and focusing on a smaller units of observation.

3.2 Reflection

Although the models, both explanatory and predictive, are invalid due to the assumptions required in the Difference in differences and MLR not being upheld, the results were still

relevant to developing an understanding of factors that impact subway ridership in a large city.

We see that the weekend has a large negative association in daily ridership. It is also interesting that whether or not it is raining has little predictive power. Subway ridership is varying metric dependent on many socioeconomic and temporal aspects.

Understanding station level or borough wide trends in ridership is useful in urban planning contexts. In anticipation of policy changes, expansions, or economic changes, these results could inform decisions on resource allocation or further research focus.

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DATA SOURCES

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MTA Subway Hourly Ridership: 2020-2024.

[7] Data.NYC.gov, https://data.ny.gov/d/5wq4-mkjj.
MTA Subway Hourly Ridership: Beginning 2025.

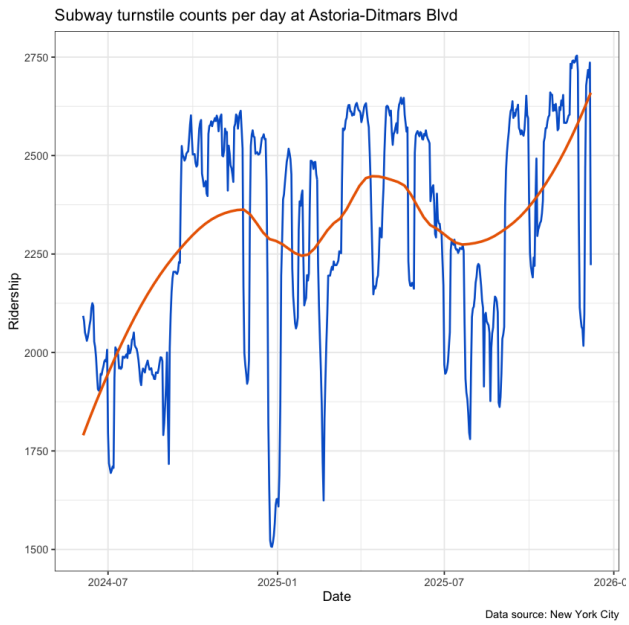
[8] NOAA. https://www.ncdc.noaa.gov/cdo-web/. JFK Airport Weather Data.

A Appendix

A.1 MLR

A.1.1 Astoria Regression

Figure 5: Figure Title



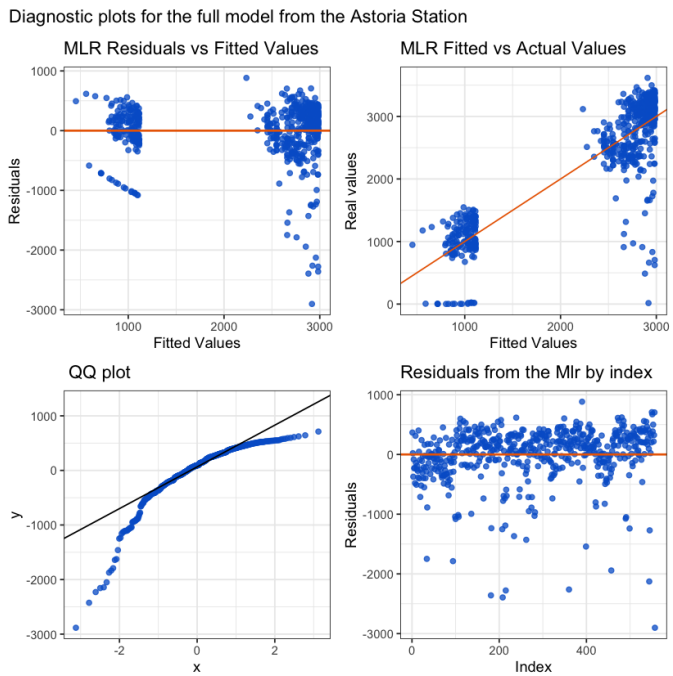
Note: Describe the figure above.

Table 4: Astoria Regression Evaluation

Model	MSPE	AIC	BIC
SLR	821603.5	7389.135	7401.429
With weekend	197261.9	6792.044	6808.437
Polynomial	183160.7	6764.780	6789.369
Full	181944.7	6760.099	6788.785

The MSPE selects the full model, as this is the one with the lowest. The AIC and BIC select the simple linear regression though. Thus, the models differ in their power for explanation and predicting new values. This could serve as evidence that I overfit the models.

Figure 6: Figure Title



Note: Graphical evidence of violations in MLR assumptions

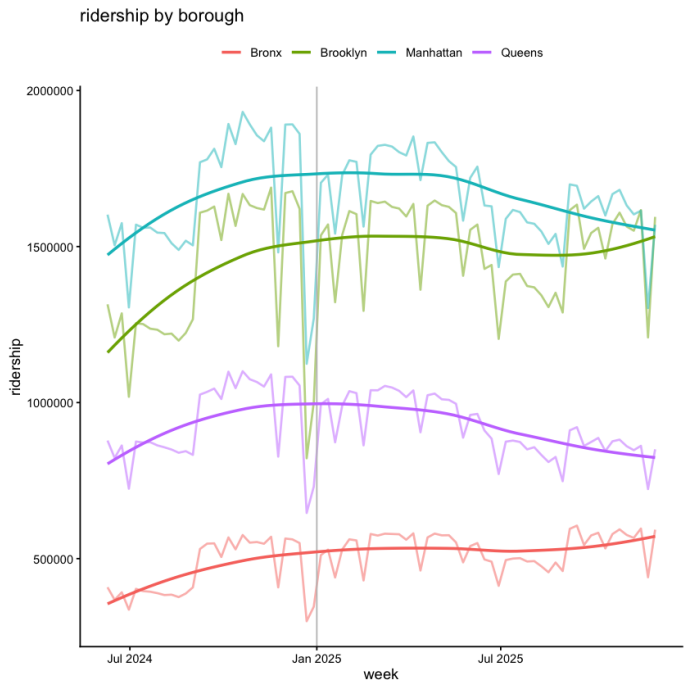
The diagnostic plots for the full regression (model 3) completed solely on data from the Astoria subway station shows that more than one assumption is violated, to a high extent.

The gap in the points on the MLR residuals vs fitted values show that the model is mispecified. Additionally, the larger variance in the cluster on the right shows that there is not constant variance in the errors.

Normality is also largely violated, as seen with the QQ plot and the Shapiro-Wilks test I had run.

A.1.2 Borough Regression

Figure 7: Ridership time series smoothed

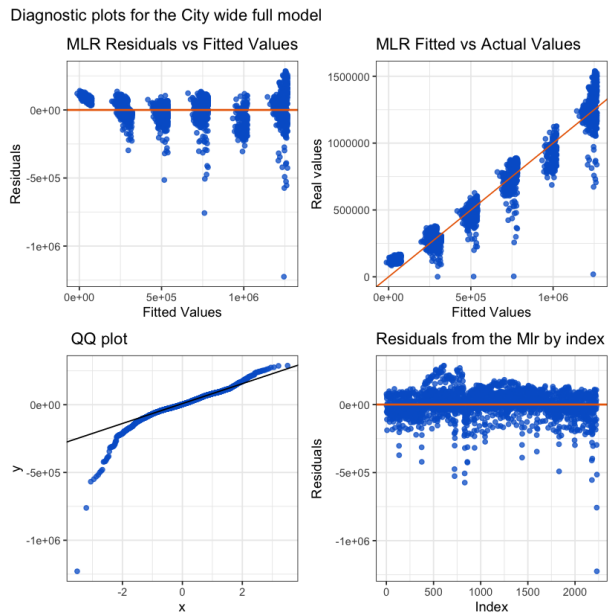


Note: The boroughs have similar spikes at the same times, but it appears that Brooklyn has more intense downward spikes than the Bronx, unadjusted

Durbin–Watson test (mod3): DW $= 0.99115$, $p < 2.2 \times 10^{-16}$;
alternative: true autocorrelation is greater than 0.

The Durbin-Watson Test reveals there there is autocorrelation between the errors. The high p value shows we reject the null hypothesis that the errors are not correlated.

Figure 8: Figure Title



The diagnostic plots for the full regression (model 4) completed city wide data also shows that the assumptions do not hold.

The gap in the points on the MLR residuals vs fitted values for this one also shows that the model is mispecified.

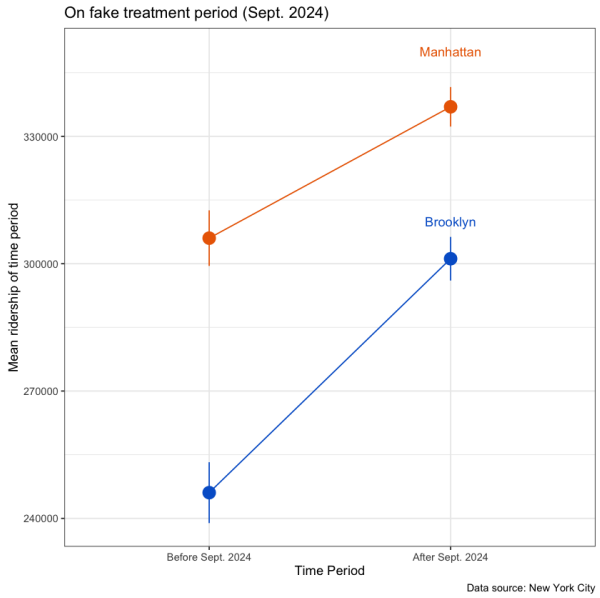
The fitted vs actual is interesting as many dots fall on the line, indicating that the real and estimated values are similar overall. Points at y=0 could indicate a data quality issue.

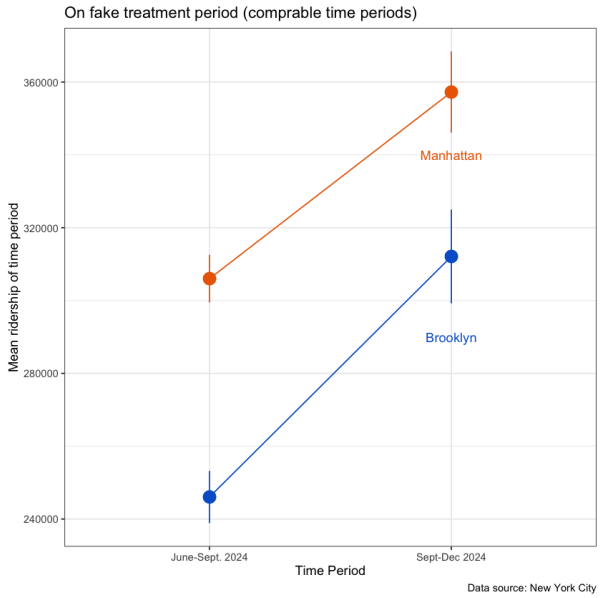
Normality is also largely violated here as well, as seen with the skew at the tails of the QQ plot.

A.2 Differences in Difference

A.3 Diff-in-diff placebo visual test results

All the placebo tests, displayed in the form of daily ridership mean over the time. These all corroborate with the result to argue that the treatment impact between Manhattan and Brooklen, before and after treatment, is not causally meaningful, as similar effects are seen with before and after arbitrary days or groups.

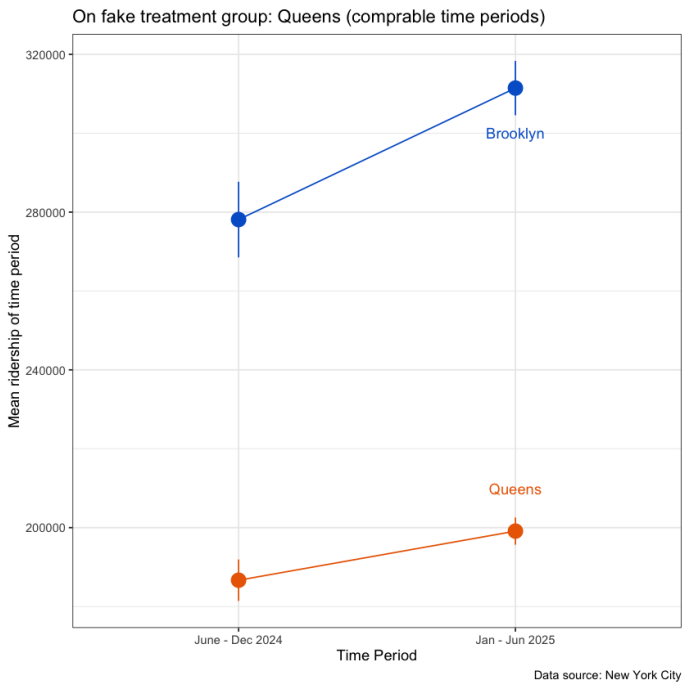




The results may be misleading exposure period is different, as before the fake treatment, only June-Sept 2024 is averaged compared to Sept 2024-June 2025. To account for this, I bound the after fake treat period to match the before.

Here I take means from the same length of times, 3 months before the fake treatment date and 3 months after. The lines are almost parallel, the Brooklyn had a greater pre-post change.

Figure 9: Figure Title



Note: Describe the figure above.

For this placebo test, I assign Queens as the fake treatment group. It also appears that Brooklyn had a greater change in ridership mean from before and after.